

Question Paper

Exam Date & Time: 24-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025
AI in Cyber Security [IT_4004]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Analyze the differences in performance between **supervised** and **unsupervised** learning approaches when applied to anomaly detection in cybersecurity. Which approach is more effective under zero-day attack conditions, and why? (5)
- A)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) Differentiate between **threats**, **vulnerabilities**, and **attacks** in the context of cybersecurity. Provide a suitable example for each and explain how they are interrelated in a real-world cyber incident. (5)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2) **Problem – Malware Behaviour Detection** (5)

A)

Sample File Access Network Activity Malware

1	High	Yes	Yes
2	Low	No	No
3	High	No	Yes
4	Low	No	No
5	High	Yes	Yes
6	Low	Yes	No

I. Calculate the Entropy of the dataset (target: Malware).

Use the formula:

$$H(S) = -p_{Yes} \log_2 p_{Yes} - p_{No} \log_2 p_{No}$$

II. Compute the Information Gain for:

1. File Access
2. Network Activity

$$IG(S, A) = H(S) - \sum_{v \in A} \frac{|S_v|}{|S|} H(S_v)$$

III. Decide which attribute becomes the Root Node.

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Critically analyze the **case study of ransomware behaviour classification** using machine learning. Discuss how real-time monitoring and feature engineering contribute to early detection. (5)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 3) Suppose a cybersecurity analyst is tasked with identifying ransomware attacks from system log data. Explain how the K-Nearest Neighbors (KNN) classification algorithm can be applied to detect and classify ransomware activities. Describe the steps involved in feature extraction, distance computation, model training, and decision-making based on the nearest neighbors. (5)
- A)

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Describe how an **Autoencoder** can be trained for **network anomaly detection**. Explain how reconstruction error is used to distinguish between **normal** and **malicious** network traffic. (5)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Discuss how **RNNs** and **LSTMs** capture sequential patterns in network traffic to detect **brute-force** or **port-scan attacks**. (5)

- A) Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Deep Learning models can automatically extract latent patterns from high-dimensional network traffic data. Analyze how Convolutional Neural Networks (CNNs) can be used to detect packet anomalies by treating packet headers as structured data or visual representations. (5)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 5) Define an **adversarial attack** in the context of machine learning. Using an example, explain how a small **perturbation or noise** added to an input (e.g., an image) can mislead a deep neural network classifier. (5)
- A)

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Illustrate with a labelled diagram the **taxonomy of adversarial attacks** based on attacker knowledge (White-box, Black-box, and Gray-box). Explain how the **attacker's access to the model** affects the type of attack used (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 IT_4004:AI in Cyber Security

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	15
3	3	CO 3	15
4	4	CO 4	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 IT_4004:AI in Cyber Security

No of

#	Level	Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	5	25(50%)
4	Analysis	5	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	10	50 (100%)